# From Ancient to Modern Italian

Ejaz Ahmed and Archit Rastogi

Università degli studi di Roma la Sapienza

## Abstract

Translating medieval Italian (12th - 15th century) into modern Italian is difficult due to major changes in vocabulary, grammar, and meaning, and because very little parallel data exists. We propose a hybrid approach that combines GPT-4.1 synthetic data, domain-specific fine-tuning, and data-free model merging using RegMean with SNR-guided layer selection. We train both linguistic and time-period expert models and evaluate them with standard metrics and an LLM-as-a-Judge. Our results show that RegMean/SNR-merged models outperform single experts and offer an efficient alternative to large LLMs, making selective merging effective for low-resource historical translation.

## 1 Introduction

Translating archaic Italian into modern Italian is challenging. Over the centuries, the language has changed in vocabulary, grammar, and meaning, and there are very few aligned datasets that pair old and modern Italian. This makes the task difficult both linguistically and from a data perspective.

To address this problem, we combine three components: (i) synthetic data generated with large LLMs, (ii) fine-tuning expert models that focus on specific linguistic or historical aspects, and (iii) efficient model merging using RegMean together with SNR-guided layer selection. Our goal is to create a single unified translator that benefits from the strengths of each expert model while avoiding catastrophic forgetting.

We also compare our system to strong generative models such as GPT-4.1 and Gemini 2.0, which serve as upper-bound references. All models are evaluated using standard translation metrics and an LLM-as-a-Judge framework that scores faithfulness, fluency, style, and overall quality.

## 2 Background and Related Work

### 2.1 Model Merging in NLP

Model merging aims to combine multiple fine-tuned models without retraining or needing access to the original datasets. Existing approaches such as weight averaging, task arithmetic, and Fisher-weighted merging, work well when the expert models are similar. However, their performance tends to drop when the domains differ significantly, often weakening the specialized knowledge learned by each expert.

### 2.2 RegMean: Regression-Based Model Merging

RegMean merges models by solving a closed-form regression using the Gram matrices of each layer:

$$W_M = \Big(\sum_i X_i^\top X_i\Big)^{-1} \sum_i X_i^\top X_i W_i.$$

This method aligns the internal representations based on activation statistics. It helps keep the important fine-tuned features while reducing noise. RegMean is data free, and is effective when merging same but specialized expert models.

### 2.3 Spectrum-Driven SNR Layer Selection

Spectrum-based SNR estimates how much meaningful information each layer contains by analyzing the eigenvalue ratios of its activation covariance. High-SNR layers hold strong and informative features, while low-SNR layers mostly contain noise. By applying RegMean only to the high-SNR layers, we can achieve almost the same merging performance while greatly reducing computational cost.

## 3 Dataset and Problem Formulation

### 3.1 Archaic to Modern Italian Translation Task

The goal of this work is to translate medieval Italian (12th - 15th century) into modern Italian. The two language forms differ significantly in morphology, syntax, vocabulary, and idiomatic use. Because aligned datasets are extremely limited, we rely on synthetic data and specialized expert models to support training.

### 3.2 Approach-1: Lexical, Syntactic, and Semantic Specialization

In the first approach, we train three linguistic experts: a lexical expert that focuses on old vocabulary and word substitutions, a syntactic expert that handles changes in grammar and word order, and a semantic expert that models meaning, idioms, and pragmatic shifts. Each expert is trained on synthetic data tailored to its specific focus. This produces strong specialization, but each model struggles to generalize outside its own domain.

### 3.3 Approach-2: Early, Middle, and Late Medieval Periods

The second approach trains three temporal experts, each dedicated to a different historical period: early (around 1260), middle (around 1310), and late (around 1360) medieval Italian. This allows the models to capture how the language changed over time. These temporal experts complement the linguistic experts by modeling natural historical evolution.

### 3.4 Real Historical Dataset

Instead of large historical corpora, we test our models on a small real dataset provided as part of the project. This dataset contains only ancient Italian sentences, while their modern versions were generated using an LLM. We then evaluate all model translations using an LLM-as-a-Judge setup based on GPT-4.1.

## 4 Methodology

### 4.1 Synthetic Data Generation

Since no large parallel dataset exists for translating archaic Italian into modern Italian, we create our own synthetic (*source*, *target*) sentence pairs. We use controlled GPT-4.1 prompts designed for each linguistic or historical domain. The generated data is then converted into JSON format.

### 4.2 Fine-Tuning Pipeline

All experts are fine-tuned on top of the same base model, `mistralai/Mistral-7B-Instruct-v0.2`, using LoRA adapters. In Approach-1, each expert focuses on a different linguistic aspect: vocabulary changes, syntactic structure, or semantic meaning. In Approach-2, each expert specializes in a specific medieval time period: early, middle, or late. All experts are trained under the same supervised setup to make comparisons consistent.

### 4.3 RegMean Merging

To combine the experts without using any training data, we use RegMean. This method merges the model parameters by solving a closed-form regression based on activation Gram matrices:

$$W_M = \Big(\sum_i X_i^\top X_i\Big)^{-1} \sum_i X_i^\top X_i W_i.$$

This helps align the internal representations of the experts, keeping useful fine-tuned information while reducing noise. Parameters outside the transformer layers are merged by simple averaging.

### 4.4 SNR-Guided Selective Layer Fusion

To make merging more efficient, we apply an SNR-based selection strategy inspired by Spectrum analysis. For each layer, we estimate an approximate SNR using the ratio between the largest eigenvalue and the average spectral trace of its activation covariance. Only the layers with the highest SNR scores are merged using RegMean, while the rest are taken from a selected reference expert. This preserves most of the benefits of full RegMean merging at a lower computational cost.

### 4.5 Unified Merging

To combine both linguistic experts (Approach-1) and temporal experts (Approach-2), we apply RegMean with SNR filtering, across all expert models. The final unified model captures a wide range of archaic styles and time periods, offering a compact and efficient solution that sits between small fine-tuned experts and large generative LLMs.

## 5 Evaluation Setup

### 5.1 Classical Metrics

We evaluate all models using standard translation metrics: BLEU, chrF++, ROUGE-L, METEOR, BERTScore, and Token-F1. These metrics measure different aspects of translation quality, including word overlap, morphological accuracy, semantic similarity, and structural alignment. All scores are computed across seven datasets that cover lexical, syntactic, semantic, temporal, and real historical variations.

### 5.2 LLM-as-a-Judge Rubrics

To capture qualitative aspects that classical metrics may miss, we use GPT-4.1 as an LLM-based evaluator. It assigns scores from 1 to 5 for four dimensions: Faithfulness, Fluency, Style, and Overall Quality. For each dataset, we report the mean and standard deviation, providing a more human-centered evaluation of clarity, correctness, and naturalness.

### 5.3 Generative Baseline Models

As strong baselines, we include two high-performing instruction-tuned LLMs: GPT-4.1 and Gemini 2.0. We test them in both zero-shot and few-shot settings. Their translations are evaluated with the same classical metrics and the LLM-as-a-Judge framework, offering upper-bound reference points for comparing our fine-tuned and merged models.

## 6 Results

We summarize the main results for the fine-tuned expert models, the RegMean/SNR merged models, and the strong generative baselines (GPT-4.1 and Gemini 2.0). Full detailed tables are reported in the Appendix.

### 6.1 Fine-Tuned Experts

The expert models show clear specialization. In Approach-1, the lexical, syntactic, and semantic experts perform well only within their own domains. In contrast, the temporal experts from Approach-2 generalize better across datasets. Among them, the A2-Late model is the strongest, while the A1-Semantic model performs the weakest. These differences highlight strong domain fragmentation and show why model merging is needed.

### 6.2 Merged Models (RegMean + SNR)

When averaging BLEU and chrF scores across all datasets, the RegMean-merged models consistently outperform the individual experts. Both Approach-1 and the unified merges benefit the most from merging. SNR-guided merging reaches similar performance while being more efficient. However, even the

best merged models still fall short of the performance achieved by large GPT-scale models.

### 6.3 Generative Baselines

GPT-4.1 achieves the best overall results, with BLEU scores around 50 - 52 and chrF around 72, even without any fine-tuning. Few-shot prompting gives small additional improvements. Gemini 2.0 is competitive but generally weaker, especially in zero-shot mode, although few-shot prompting helps make its output more stable.

### 6.4 LLM-as-a-Judge Evaluation

The LLM-as-a-Judge results (faithfulness, fluency, style, and overall quality) support the same trends seen in classical metrics. GPT-4.1 receives near-human ratings (4.6 - 5.0). Gemini scores slightly lower (4.3 - 4.9). The merged models form a middle tier (around 3.0 - 3.3), while the individual experts score the lowest (2.3 - 3.1). Merging noticeably improves fluency and style compared to isolated experts.

### 6.5 Cross-Family Comparison

Large LLMs clearly perform the best across both quantitative metrics and qualitative evaluations. RegMean/SNR merged models show meaningful improvements over individual experts and generalize more reliably, placing them in a solid mid-range tier. Expert models alone work well only within their narrow domains. SNR-guided selective merging preserves most of the benefits of full RegMean while reducing computational cost.

## 7 Discussion

### 7.1 A2 vs. A1 Experts

The A2 temporal experts perform better than the A1 linguistic experts because the historical development of Italian is gradual and follows a clear timeline. Specializing models by historical period therefore creates more coherent and generalizable expertise. In contrast, Approach-1 divides the problem into very narrow lexical, syntactic, and semantic components. These experts become highly specialized, but they struggle to generalize outside their limited focus areas.

### 7.2 Impact of RegMean

RegMean improves performance by aligning the expert models using activation-level regression. This helps preserve stable and informative feature spaces while merging their complementary knowledge. As a result, the merged models can capture lexical, syntactic, semantic, and temporal information at the same time. Even though these merged models are much smaller than GPT-scale LLMs, they form a strong mid-range solution.

### 7.3 Why GPT-4.1 Leads

GPT-4.1 achieves the highest performance because of its massive pretraining, its exposure to multiple languages, its instruction tuning, and its strong in-context reasoning ability. It can handle archaic grammar, vocabulary, and stylistic patterns even in zero-shot settings. This leads to near-human translation quality that small fine-tuned models cannot reach.

### 7.4 BLEU vs. LLM-Judge

BLEU is useful for measuring meaning preservation, but it does not capture fluency or natural style very well. The LLM-as-a-Judge evaluations show that models with moderate BLEU scores can still produce fluent text, while models with high BLEU may sound less natural. Using both classical metrics and LLM-based qualitative judgments gives a more complete picture of translation quality.

## 8 Conclusion and Future Work

This study shows that using synthetic data, domain-specialized fine-tuning, and RegMean/SNR merging can produce efficient models for translating archaic Italian into modern Italian. The expert models

learn their specific domains well but do not generalize widely, while the merged models provide a balanced and lightweight alternative. Although GPT-4.1 and Gemini still represent the upper bound in performance, model merging offers a practical middle ground that does not require large computational resources.

For future work, we plan to use real historical corpora, explore dynamic SNR-based layer selection, experiment with unified decoder-only translation models, and incorporate multilingual LLM-judge evaluation. These directions could help further close the gap between compact merged models and large generative LLMs.

# Appendix

This appendix reports all complete evaluation tables: fine-tuned experts, merged models, GPT/Gemini models, and LLM-as-a-Judge scores.

## A.1 Fine-Tuned Experts (Approach-1)

Table 1: A1 Experts - Full Metrics (Part 1)

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| a1_lexical_expert - a1_lexical_test | 35.27 | 60.75 | 57.06 | 0.57 | 84.89 | 57.13 |
| a1_lexical_expert - real_97_test | 21.86 | 46.11 | 43.33 | 0.45 | 79.13 | 43.08 |
| a1_syntactic_expert - a1_syntactic_test | 16.24 | 50.11 | 41.81 | 0.45 | 81.21 | 40.24 |
| a1_syntactic_expert - real_97_test | 22.11 | 48.31 | 44.20 | 0.46 | 80.94 | 45.27 |
| a1_semantic_expert - a1_semantic_test | 9.49 | 34.15 | 28.49 | 0.29 | 75.92 | 29.38 |
| a1_semantic_expert - real_97_test | 15.12 | 41.69 | 35.44 | 0.38 | 78.26 | 36.00 |

## A.1 Fine-Tuned Experts (Approach-2)

Table 2: A2 Experts - Full Metrics

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| a2_early_expert - a2_early_1260_test | 41.07 | 65.32 | 62.10 | 0.67 | 86.87 | 59.93 |
| a2_early_expert - real_97_test | 28.65 | 51.84 | 48.23 | 0.51 | 82.58 | 48.16 |
| a2_middle_expert - a2_middle_1310_test | 32.37 | 54.35 | 47.86 | 0.51 | 83.25 | 48.19 |
| a2_middle_expert - real_97_test | 30.02 | 53.35 | 52.00 | 0.53 | 83.52 | 51.25 |
| a2_late_expert - a2_late_1360_test | 57.78 | 70.92 | 71.49 | 0.75 | 90.23 | 72.81 |
| a2_late_expert - real_97_test | 18.83 | 43.83 | 37.56 | 0.39 | 78.90 | 37.99 |

## A.2 Merged Models - A1

Table 3: A1 Merged Models

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| a1_merged_lora - a1_lexical_test | 22.31 | 49.77 | 42.30 | 0.45 | 83.24 | 41.21 |
| a1_merged_lora - a1_syntactic_test | 17.47 | 49.63 | 41.64 | 0.44 | 82.86 | 40.32 |
| a1_merged_lora - a1_semantic_test | 12.14 | 34.72 | 30.30 | 0.32 | 77.33 | 31.30 |
| a1_merged_lora - real_97_test | 19.08 | 47.17 | 43.39 | 0.46 | 81.48 | 43.53 |
| a1_merged_full - a1_lexical_test | 27.46 | 55.42 | 46.92 | 0.50 | 85.49 | 48.63 |
| a1_merged_full - a1_syntactic_test | 19.87 | 51.33 | 47.06 | 0.49 | 84.44 | 45.38 |
| a1_merged_full - a1_semantic_test | 9.71 | 36.06 | 28.41 | 0.29 | 76.54 | 28.58 |
| a1_merged_full - real_97_test | 19.15 | 46.68 | 42.41 | 0.45 | 81.17 | 42.76 |

## A.2 Merged Models - A2

Table 4: A2 Merged Models

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| a2_merged_lora - a2_early_1260_test | 20.33 | 49.58 | 44.38 | 0.48 | 83.00 | 41.70 |
| a2_merged_lora - a2_middle_1310_test | 19.51 | 49.04 | 42.81 | 0.45 | 82.52 | 44.37 |
| a2_merged_lora - a2_late_1360_test | 16.74 | 48.21 | 47.80 | 0.50 | 82.45 | 44.55 |
| a2_merged_lora - real_97_test | 18.79 | 45.59 | 41.15 | 0.44 | 80.79 | 41.22 |
| a2_merged_full - a2_early_1260_test | 13.82 | 43.22 | 38.41 | 0.43 | 79.56 | 39.28 |
| a2_merged_full - a2_middle_1310_test | 14.61 | 45.41 | 37.61 | 0.38 | 80.60 | 37.07 |
| a2_merged_full - a2_late_1360_test | 17.20 | 46.19 | 41.58 | 0.44 | 82.46 | 43.49 |
| a2_merged_full - real_97_test | 19.97 | 46.33 | 42.51 | 0.46 | 80.97 | 42.97 |

## A.2 Unified Merged Model

Table 5: Unified Merged Model - Full Metrics

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| unified_merged_full - a1_lexical_test | 24.09 | 49.48 | 43.54 | 0.47 | 81.78 | 43.03 |
| unified_merged_full - a1_syntactic_test | 12.19 | 45.44 | 38.10 | 0.39 | 79.07 | 37.88 |
| unified_merged_full - a1_semantic_test | 9.89 | 35.79 | 24.19 | 0.24 | 76.04 | 21.68 |
| unified_merged_full - a2_early_1260_test | 23.70 | 47.26 | 40.85 | 0.46 | 81.72 | 39.65 |
| unified_merged_full - a2_middle_1310_test | 16.14 | 48.51 | 40.75 | 0.43 | 81.07 | 41.32 |
| unified_merged_full - a2_late_1360_test | 20.68 | 45.32 | 44.17 | 0.45 | 83.68 | 43.16 |
| unified_merged_full - real_97_test | 20.69 | 47.21 | 43.93 | 0.46 | 81.18 | 43.42 |

## A.3 GPT-4.1 Results

Table 6: GPT-4.1 Full Metrics

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| gpt4_1 - a2_middle_1310_test | 67.74 | 83.87 | 85.21 | 0.87 | 95.11 | 83.16 |
| gpt4_1 - a1_semantic_test | 28.51 | 53.65 | 56.59 | 0.55 | 84.48 | 56.21 |
| gpt4_1 - real_97_test | 69.92 | 81.91 | 85.14 | 0.85 | 95.32 | 83.84 |
| gpt4_1 - a2_early_1260_test | 70.17 | 81.82 | 83.41 | 0.85 | 94.55 | 82.27 |
| gpt4_1 - a1_lexical_test | 33.44 | 59.69 | 64.60 | 0.65 | 87.17 | 60.89 |
| gpt4_1 - a1_syntactic_test | 36.17 | 64.16 | 58.53 | 0.63 | 88.43 | 62.67 |
| gpt4_1 - a2_late_1360_test | 61.19 | 76.83 | 74.95 | 0.74 | 92.52 | 71.09 |

## A.3 GPT-4.1 Few-Shot Results

Table 7: GPT-4.1 Few-Shot Metrics

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| gpt4_1_fewshot - a1_syntactic_test | 39.39 | 65.00 | 59.02 | 0.64 | 88.91 | 64.34 |
| gpt4_1_fewshot - a2_early_1260_test | 69.40 | 81.68 | 83.64 | 0.84 | 94.24 | 82.10 |
| gpt4_1_fewshot - a2_late_1360_test | 61.31 | 78.95 | 79.86 | 0.80 | 94.09 | 75.77 |
| gpt4_1_fewshot - real_97_test | 68.42 | 81.24 | 83.28 | 0.85 | 94.48 | 82.61 |
| gpt4_1_fewshot - a1_lexical_test | 31.18 | 59.29 | 64.99 | 0.67 | 87.77 | 61.66 |
| gpt4_1_fewshot - a1_semantic_test | 27.23 | 51.66 | 49.47 | 0.50 | 84.04 | 50.55 |
| gpt4_1_fewshot - a2_middle_1310_test | 66.42 | 83.33 | 84.16 | 0.87 | 95.09 | 84.37 |

## A.3 Gemini 2.5 Results

Table 8: Gemini 2.5 Full Metrics

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| gemini_2_5 - a2_early_1260_test | 35.33 | 59.81 | 61.03 | 0.63 | 87.74 | 60.37 |
| gemini_2_5 - a2_late_1360_test | 32.87 | 59.10 | 60.98 | 0.60 | 88.69 | 60.46 |
| gemini_2_5 - a2_middle_1310_test | 43.45 | 67.24 | 69.05 | 0.72 | 89.68 | 68.38 |
| gemini_2_5 - a1_lexical_test | 24.17 | 55.21 | 63.64 | 0.64 | 86.15 | 60.00 |
| gemini_2_5 - a1_semantic_test | 27.64 | 51.58 | 52.10 | 0.51 | 84.05 | 48.99 |
| gemini_2_5 - real_97_test | 42.34 | 64.28 | 68.10 | 0.68 | 89.50 | 65.75 |
| gemini_2_5 - a1_syntactic_test | 30.97 | 56.52 | 55.53 | 0.63 | 87.32 | 59.36 |

## A.3 Gemini 2.5 Few-Shot Results

Table 9: Gemini 2.5 Few-Shot Metrics

| Model / Dataset | BLEU | chrF | ROUGE-L | METEOR | BERT | TokF1 |
|---|---|---|---|---|---|---|
| gemini2_5_fewshot - a2_early_1260_test | 59.99 | 74.89 | 75.45 | 0.78 | 94.17 | 73.60 |
| gemini2_5_fewshot - a2_middle_1310_test | 42.28 | 68.89 | 67.68 | 0.70 | 90.33 | 66.77 |
| gemini2_5_fewshot - a1_lexical_test | 27.91 | 54.84 | 64.23 | 0.66 | 86.82 | 61.01 |
| gemini2_5_fewshot - real_97_test | 44.73 | 65.97 | 68.82 | 0.70 | 90.42 | 67.45 |
| gemini2_5_fewshot - a1_semantic_test | 27.70 | 51.99 | 47.40 | 0.48 | 84.44 | 45.80 |
| gemini2_5_fewshot - a2_late_1360_test | 35.40 | 61.05 | 62.76 | 0.59 | 89.10 | 59.66 |
| gemini2_5_fewshot - a1_syntactic_test | 37.54 | 63.91 | 59.64 | 0.68 | 88.69 | 63.32 |

## A.4 LLM-as-a-Judge - GPT-4.1 Few-Shot

Table 10: Judge Scores - GPT-4.1 Few-Shot

| Model / Dataset | Faith | Fluency | Style | Overall |
|---|---|---|---|---|
| gpt4_1_fewshot - a2_middle_1310_test | 4.75±0.43 | 5.00 | 4.83±0.37 | 4.75±0.43 |
| gpt4_1_fewshot - a1_syntactic_test | 4.17±0.37 | 5.00 | 4.50±0.50 | 4.33±0.47 |
| gpt4_1_fewshot - a1_lexical_test | 4.60±0.49 | 5.00 | 4.80±0.40 | 4.80±0.40 |
| gpt4_1_fewshot - a2_early_1260_test | 4.70±0.46 | 5.00 | 4.90±0.30 | 4.80±0.40 |
| gpt4_1_fewshot - real_97_test | 4.64±0.59 | 5.00 | 4.84±0.36 | 4.75±0.52 |
| gpt4_1_fewshot - a2_late_1360_test | 4.50±0.50 | 5.00 | 4.67±0.47 | 4.67±0.47 |
| gpt4_1_fewshot - a1_semantic_test | 4.40±0.49 | 5.00 | 4.80±0.40 | 4.80±0.40 |

## A.4 LLM-as-a-Judge - Gemini 2.5 Few-Shot

Table 11: Judge Scores - Gemini 2.5 Few-Shot

| Model / Dataset | Faith | Fluency | Style | Overall |
|---|---|---|---|---|
| gemini2_5_fewshot - a2_middle_1310_test | 4.00±0.63 | 5.00 | 4.50±0.50 | 4.30±0.46 |
| gemini2_5_fewshot - a1_syntactic_test | 4.50±0.50 | 5.00 | 5.00 | 5.00 |
| gemini2_5_fewshot - a1_lexical_test | 4.40±0.49 | 5.00 | 4.80±0.40 | 4.60±0.49 |
| gemini2_5_fewshot - a2_early_1260_test | 4.70±0.46 | 5.00 | 4.90±0.30 | 4.90±0.30 |
| gemini2_5_fewshot - real_97_test | 4.41±0.55 | 4.98±0.14 | 4.64±0.48 | 4.56±0.52 |
| gemini2_5_fewshot - a1_semantic_test | 4.60±0.49 | 5.00 | 5.00 | 4.80±0.40 |

## A.4 LLM-as-a-Judge - Fine-Tuned Experts

Table 12: Judge Scores - Experts

| Model / Dataset | Faith | Fluency | Style | Overall |
|---|---|---|---|---|
| a1_lexical - a1_lexical_test | 2.60±0.49 | 2.80±0.40 | 2.80±0.40 | 2.60±0.49 |
| a1_lexical - real_97_test | 2.61±0.84 | 3.06±0.95 | 2.68±0.77 | 2.55±0.84 |
| a1_syntactic - a1_syntactic | 2.33±0.47 | 2.50±0.76 | 2.17±0.37 | 2.33±0.47 |
| a1_semantic - a1_semantic | 2.80±0.40 | 3.40±1.20 | 3.00±0.89 | 2.60±0.49 |
| a1_semantic - real_97_test | 2.35±0.75 | 3.60±1.04 | 2.93±0.80 | 2.50±0.73 |
| a2_middle - a2_middle | 3.08±0.95 | 3.67±1.31 | 3.17±1.14 | 2.83±1.07 |
| a2_middle - real_97_test | 3.09±0.91 | 4.01±1.05 | 3.41±0.93 | 3.21±0.92 |
| a2_late - a2_late | 3.67±0.74 | 4.33±0.74 | 3.50±0.96 | 3.67±0.74 |
| a2_late - real_97_test | 2.31±0.69 | 2.91±0.94 | 2.47±0.69 | 2.36±0.69 |

## A.4 LLM-as-a-Judge - Merged Models

Table 13: Judge Scores - Merged Models

| Model / Dataset | Faith | Fluency | Style | Overall |
|---|---|---|---|---|
| a1_merged_full - a1_syntactic_test | 3.00 | 4.17±0.90 | 3.67±0.74 | 3.50±0.50 |
| a2_merged_lora - a2_middle_test | 3.08±0.64 | 3.92±0.86 | 3.33±0.62 | 3.17±0.69 |
| a2_merged_full - a2_late_test | 2.83±0.37 | 4.00±0.82 | 3.33±0.74 | 2.83±0.37 |
| unified_merged - a2_middle | 2.75±0.72 | 3.75±0.72 | 3.42±0.64 | 3.25±0.72 |
| unified_merged - real_97_test | 2.88±0.71 | 3.76±0.87 | 3.14±0.70 | 2.98±0.75 |

## A.5 Prompt Templates

This section reports all prompt formulations used across the pipeline: synthetic data creation, LoRA fine-tuning, GPT/Gemini translation baselines, and LLM-as-a-Judge evaluation. Prompts are reported verbatim (up to light formatting) to ensure full reproducibility.

### A.5.1 Synthetic Data Generation - Approach 1 (Linguistic Experts)

**Approach 1 Prompt (Lexical / Syntactic / Semantic)**

You are an expert in historical Italian linguistics specializing in modernizing 12th - 15th century texts.
**Task**: Modernize the following archaic Italian sentence according to the specific scenario.
**Modernization Scenario**: {`scenario_name`}
**Scenario Objective**: {`scenario_description`}
**Original Sentence**: {`sentence`}
**Metadata**: Author = {`author`}, Period = {`date`}, Region = {`region`}
**Instructions**:

- Apply only transformations relevant to the scenario.
- Preserve meaning and semantic content.
- Ensure output is natural 2025 Italian.
- Return only the modernized sentence in the `target` field.
- Do not include explanations.

**CRITICAL**: The `target` field must contain only the modernized sentence.

### A.5.2 Synthetic Data Generation - Approach 2 (Temporal Experts)

**Approach 2 Prompt (Early / Middle / Late Period)**

You are an expert in diachronic Italian linguistics specializing in {`time_range`}.
**Task**: Modernize the following archaic sentence from {`time_range`} into natural 2025 Italian.
**Period Context**: Linguistic properties of this era include {`linguistic_features`}.
**Original Sentence**: {`sentence`}
**Instructions**:

- Recognize period-specific linguistic phenomena.
- Preserve meaning exactly.
- Produce fluent contemporary Italian.
- Output only the modern sentence in `target` (no commentary).

**CRITICAL**: Only the modern sentence is allowed in the output.

### A.5.3 LoRA Fine-Tuning Instruction Format

**Supervised Instruction Format (All LoRA Experts)**

**Istruzione:**
Modernizza la seguente frase italiana arcaica in italiano contemporaneo.

**Frase arcaica:**
{`source`}

**Frase moderna:**
{`target`}

### A.5.4 Evaluation Translation Prompt (Inference Phase)

Inference Prompt (Evaluation Stage)

**Istruzione:**
Modernizza la seguente frase italiana arcaica in italiano contemporaneo.

**Frase arcaica:**
{`text`}

**Frase moderna:**

### A.5.5 GPT-4.1 Zero-Shot Translation Prompt

GPT-4.1 Zero-Shot Prompt

You are a professional Italian linguist specialized in transforming archaic (13th–15th century) Italian into natural 2025 Italian.

**Task**
Translate the following archaic Italian sentence into fluent modern Italian, preserving meaning and stylistic naturalness.

**Archaic Sentence**
{`text`}

**Output Rules**

- Keep the meaning identical.
- Use modern vocabulary and syntax.
- Output only the translation text (no explanations).

### A.5.6 GPT-4.1 Few-Shot Translation Prompt (with Examples)

GPT-4.1 Few-Shot Prompt

You are a professional Italian linguist specialized in transforming archaic (13th–15th century) Italian into natural 2025 Italian.
Learn from the examples below and translate the following archaic sentence into fluent modern Italian while preserving meaning and tone.

**Example 1**
`"source": "sono due già non in una carne, ma in uno spirito, cioè Iddio, e l' anima. Onde in altro luogo dice S. Paolo: Chi s' accosta a Dio è uno spirito",`
`"target": "sono due già non in una carne, ma in uno spirito, cioè Dio e l'anima. Onde in un altro luogo dice San Paolo: Chi si accosta a Dio è uno spirito"`

**Example 2**
`"source": "Altressì uno amante chiamando merzé alla sua donna dice parole e ragioni molte, et ella si difende in suo dire.",`
`"target": "Allo stesso modo, un uomo innamorato, mentre chiede misericordia alla donna che ama, le rivolge molte parole e argomentazioni, ed essa risponde difendendo la propria posizione con le sue parole."`

**Sentence to translate**
{`text`}

**Output format (strict JSON)**
`{"translation": "<modern translation>"}`

### A.5.7 Gemini 2.5 Zero-Shot Translation Prompt

**Gemini 2.5 Zero-Shot Prompt**

The Gemini 2.5 Zero-Shot translation uses the **same prompt structure as the GPT-4.1 Zero-Shot** translation prompt described in Section A.5.5.
The only change is the model endpoint, while the instruction, task description, and output formatting requirements remain identical.

**Archaic sentence**
{`text`}

**Output**
Only the modern Italian translation (no explanations), returned in the `translation` JSON field.

### A.5.8 Gemini 2.5 Few-Shot Translation Prompt

**Gemini 2.5 Few-Shot Prompt**

The Gemini 2.5 Few-Shot translation uses the **same prompt structure and the same five few-shot examples as the GPT-4.1 Few-Shot** translation prompt presented in Section A.5.6. No modifications were introduced to ensure a fair comparison across LLM baselines.

**Sentence to translate**
{`text`}

**Output format (strict JSON)**
{`"translation": "<modern translation>"`}

### A.5.9 LLM-as-a-Judge Prompt and Rubrics (GPT-4.1)

**LLM-as-a-Judge Evaluation Prompt**

You are a bilingual Italian linguist specialized in translating and evaluating texts from medieval Italian (13th–15th century) to modern Italian.
Evaluate the following translation according to the rubrics below. Use the reference translation only as a *semantic anchor* (not as a gold standard). Do not paraphrase or retranslate; only judge the given model output.
Provide feedback (at most 50 words) and a score from 1 to 5 for each criterion.

**Archaic Sentence**
{`source`}

**Model Translation**
{`prediction`}

**Reference Translation (semantic anchor)**
{`reference`}

**Rubrics**
**Faithfulness** – Evaluate meaning preservation and modernization accuracy.
5: Excellent (meaning fully preserved and properly modernized).
4: Good (minor nuance loss or slightly literal).
3: Fair (understandable but partial misinterpretation).
2: Poor (major omission or distortion).
1: Fail (meaning lost or wrong).

**Fluency** – Evaluate grammar, syntax, and comprehensibility.
5: Excellent (flawless, idiomatic modern Italian).
4: Good (minor slips but very readable).
3: Fair (awkward phrasing or rigid syntax).
2: Poor (frequent grammatical errors).
1: Fail (broken or barely readable).

**Style** – Evaluate naturalness and human-likeness of the translation.
5: Excellent (feels human, elegant, and modern).
4: Good (smooth but slightly mechanical).
3: Fair (understandable but stiff).
2: Poor (robotic tone or phrasing).
1: Fail (stylistically incoherent).

**Overall** – Holistic quality combining all above criteria.
5: Excellent (faithful, fluent, and natural).
4: Good (minor issues not affecting comprehension).
3: Fair (acceptable but mechanical).
2: Poor (noticeable problems).
1: Fail (unusable translation).

**Required JSON Output**

```
"Faithfulness": "feedback": "...", "score": X
, "Fluency": "feedback": "...", "score": X
, "Style": "feedback": "...", "score": X ,
"Overall": "feedback": "...", "score": X
```